CASE STUDY

on

AI for Tele-Dermatology

Skin Disease Classification & Lesion Segmentation

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1. Introduction

Skin diseases represent a significant global health burden, affecting approximately one in three people worldwide. Early and accurate diagnosis is critical for effective treatment, particularly for conditions such as melanoma, where five-year survival rates drop dramatically with late detection. Traditional dermatological diagnosis relies on specialist consultation, which is inaccessible in many regions due to geographic, economic, and resource constraints.

Tele-dermatology offers a transformative solution by enabling remote diagnosis and monitoring of skin conditions using images captured through mobile devices. However, real-world tele-dermatology presents unique challenges: images are captured under varying illumination conditions, across diverse skin tones, with different devices, and often with varying levels of image quality and noise. These factors introduce significant variability that can degrade the performance of automated diagnostic models.

Computer vision and deep learning have emerged as powerful tools for automated skin lesion analysis. Two complementary tasks are central to this domain. Classification supports clinical decision-making by categorizing lesions into disease types, while segmentation enables precise pixel-level lesion delineation, which is critical for measuring lesion size, shape, and boundary characteristics — all key clinical parameters. This two-week case study investigates the design, training, and critical evaluation of deep learning models for both tasks under conditions that reflect real-world tele-dermatology deployments, with a particular focus on robustness to brightness variation and skin-tone diversity.

2. Problem Statement

Despite significant advances in AI-based dermatology, two critical gaps remain. First, most published models are evaluated on well-controlled benchmark datasets that do not reflect the variability encountered during actual tele-dermatology deployments. Second, existing models have demonstrated systematic performance disparities across skin-tone groups, raising serious concerns about equity and bias in clinical AI applications.

This case study addresses the following problem: Can deep learning models accurately classify skin disease types and precisely segment lesion boundaries when presented with images that vary in brightness — simulating different lighting conditions — and across different skin-tone groups? Which task, classification or segmentation, is more sensitive to these real-world perturbations, and what are the implications for clinical deployment?

The academic impact of this work lies in developing a robust experimental framework for evaluating model fairness and robustness, contributing to the growing body of literature on bias-aware medical AI. The practical impact addresses the need for reliable automated tools in tele-dermatology settings, particularly for underserved populations with limited access to specialist care.

3. Objectives of the Study

3.1 Analytical Goals

The primary analytical goals are to design, train, and evaluate a ResNet50-based multi-class skin disease classification model using the ISIC 2018 dataset, and to build and evaluate a U-Net segmentation model for pixel-level lesion delineation using matched image-mask pairs from the ISIC 2018 Task 1 dataset.

3.2 Robustness and Visualization Goals

The study aims to systematically quantify the effect of brightness variation (factors 0.3, 0.6, 1.0, 1.5, and 2.0) on both classification and segmentation performance. Additionally, the study examines performance disparities across skin-tone groups — light, medium, and dark — estimated from image luminance, with the objective of identifying potential algorithmic bias.

3.3 Decision-Support and Performance Goals

The study seeks to generate clinically relevant insights about which disease classes are most prone to misclassification, identify the predominant types of segmentation errors (over-segmentation, boundary leakage, missed lesions), and produce actionable recommendations for improving model robustness for real-world tele-dermatology deployment. Performance benchmarks include achieving accuracy above 85% for classification and a Dice score above 0.87 for segmentation.

4. Literature Review

Esteva et al. (2017) demonstrated that a CNN trained on dermoscopic images achieved dermatologist-level performance on melanoma classification, establishing the feasibility of deep learning for clinical dermatology. Subsequent works leveraging transfer learning — particularly VGG, ResNet, and Inception-based architectures — have consistently achieved high accuracy on benchmark datasets such as HAM10000 and ISIC 2018.

For segmentation, Ronneberger et al. (2015) introduced U-Net, which has since become the de-facto standard for biomedical image segmentation. The encoder-decoder architecture with skip connections is particularly effective at capturing both high-level semantic features and fine-grained spatial details necessary for accurate lesion boundary delineation. Zhou et al. (2018) extended this with U-Net++, improving performance through nested dense connections.

A critical gap in the literature is the limited evaluation of these models under real-world distribution shifts. Groh et al. (2021) specifically highlighted that dermatology AI datasets are heavily biased toward lighter skin tones, with the ISIC dataset having fewer than 5% of images from individuals with darker skin. This creates models that systematically underperform for underrepresented populations. Similarly, studies on image quality degradation have shown that brightness and contrast variations — common in mobile photography — can cause significant performance drops in models not trained with corresponding augmentations.

This case study is grounded in these findings and builds upon them by providing controlled experimental evidence of how these factors affect both classification and segmentation in a unified framework.

5. Data Description

5.1 Data Source

This study uses the ISIC 2018 Challenge Dataset, a publicly available benchmark for skin lesion analysis. Task 3 of the challenge provides 10,015 dermoscopic images with corresponding one-hot encoded ground-truth labels for seven disease categories. Task 1 provides 2,594 image-mask pairs for lesion segmentation. All images were downloaded directly from the ISIC challenge archive (isic-challenge-data.s3.amazonaws.com) and are fully open-access.

5.2 Data Structure

The Task 3 classification dataset contains 10,015 images in JPEG format, accompanied by a CSV file with one-hot encoded labels for seven disease classes. The Task 1 segmentation dataset contains 2,594 images in JPEG format with corresponding binary PNG segmentation masks. The dataset exhibits significant class imbalance, with NV comprising 66.9% of samples, while DF and VASC together account for only 2.6% — a critical consideration for model training and evaluation.

5.3 Dataset Split

For classification, the 10,015 images were split using stratified sampling: 7,010 for training (70%), 1,502 for validation (15%), and 1,503 for testing (15%). For segmentation, the 2,594 matched pairs were split 80/20: 2,075 for training and 519 for testing.

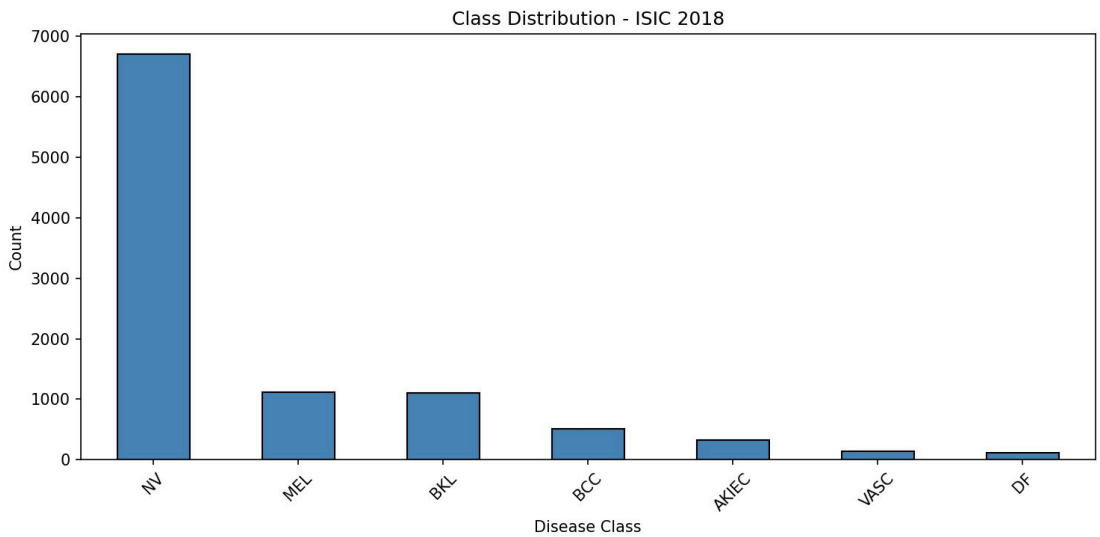

Figure 1: Class Distribution of the ISIC 2018 Task 3 Dataset (10,015 samples). NV dominates with 6,705 samples, while DF (115) and VASC (142) are severely underrepresented.

5.4 Data Dictionary

Attribute	Description	Data Type	Range / Example	Role	Notes
image	Unique image ID	String	ISIC_0024306	Identifier	Linked to .jpg file
MEL	Melanoma label	Binary (0/1)	0 or 1	Target label	Malignant
NV	Melanocytic nevus	Binary (0/1)	0 or 1	Target label	Most frequent class
BCC	Basal cell carcinoma	Binary (0/1)	0 or 1	Target label	Skin cancer

AKIEC	Actinic keratosis	Binary (0/1)	0 or 1	Target label	Pre-cancerous
BKL	Benign keratosis	Binary (0/1)	0 or 1	Target label	Benign growth
DF	Dermatofibroma	Binary (0/1)	0 or 1	Target label	Rare class
VASC	Vascular lesion	Binary (0/1)	0 or 1	Target label	Rarest class
path	Full file path to image	String (path)	/content/...jpg	Input	Derived column
skin_tone	Estimated skin tone group	Categorical	Light/Medium/Dark	Analysis	Derived via luminance

6. Methodology

6.1 Data Collection and Preprocessing

Images were downloaded directly in Google Colab from the official ISIC challenge S3 bucket. Image integrity was verified using PIL's image.verify() method to remove corrupted files. All images were resized to 224×224 pixels for classification and 256×256 pixels for segmentation. Normalization was applied using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) for classification, maintaining compatibility with pretrained ImageNet weights.

6.2 Data Augmentation

For classification training, standard augmentation was applied: random horizontal flip (p=0.5), random vertical flip (p=0.5), random rotation (±15°), and ColorJitter with brightness=0.2 and contrast=0.2. For segmentation, brightness augmentation was applied using Albumentations: RandomBrightnessContrast with brightness_limit=0.3 and contrast_limit=0.3. A separate model was trained without this augmentation to quantify its effect on segmentation robustness.

6.3 Classification Model Architecture

A ResNet50 model pretrained on ImageNet was employed as the backbone for classification. The final fully connected layer was replaced with a new linear layer mapping 2,048 features to 7 output classes. The model was trained for 15 epochs using the Adam optimizer (learning rate 1e-4) with a StepLR scheduler (step_size=5, gamma=0.5) and cross-entropy loss. Batch size was set to 16.

6.4 Segmentation Model Architecture

A U-Net model with a ResNet34 encoder pretrained on ImageNet was constructed using the segmentation-models-pytorch (SMP) library. The model outputs a single-channel binary mask. Training used a combined loss function of Dice Loss and Binary Cross-Entropy (BCE) Loss, which improves boundary precision. The model was trained for 10 epochs using the Adam optimizer (learning rate 1e-4) with batch size 8.

6.5 Skin Tone Estimation

Skin tone groups were estimated by computing the mean luminance of each image using grayscale conversion. Images with mean luminance below 100 were classified as Dark, between 100 and 160 as Medium, and above 160 as Light. This proxy method, while approximate, provides a practical basis for analyzing model performance disparities across skin tone groups in the absence of explicit skin tone annotations in the ISIC dataset.

6.6 Brightness Robustness Analysis

To evaluate classification robustness, the trained ResNet50 model was evaluated on the test set with five brightness levels applied at inference time: factors of 0.3, 0.6, 1.0, 1.5, and 2.0 using torchvision ColorJitter. For segmentation, two U-Net models were trained and compared — one with brightness augmentation and one without — and evaluated on the same test set.

7. Exploratory Data Analysis (EDA)

The ISIC 2018 dataset is heavily imbalanced. Melanocytic nevi (NV) accounts for 6,705 samples (66.9%), while dermatofibroma (DF) and vascular lesions (VASC) together represent only 2.6% of the dataset. This imbalance has direct implications for model performance, as the classifier naturally biases toward the majority NV class. The confusion matrix confirms this: NV achieves the highest recall (0.95) while low-frequency classes like MEL suffer significantly (recall 0.57).

The skin tone distribution in the test set reveals another critical imbalance: 624 light-toned samples (41.5%), 876 medium-toned samples (58.3%), and only 3 dark-toned samples (0.2%). While the extreme scarcity of dark-toned images limits statistical conclusions, the performance gap (accuracy 0.8782 for light vs. 0.6667 for dark) is directionally consistent with documented biases in dermatology AI literature.

Training curves show healthy convergence: training loss decreases consistently from 0.7395 to 0.0602 over 15 epochs, while validation accuracy improves from 0.7790 to 0.8782 with no signs of overfitting within the training window.

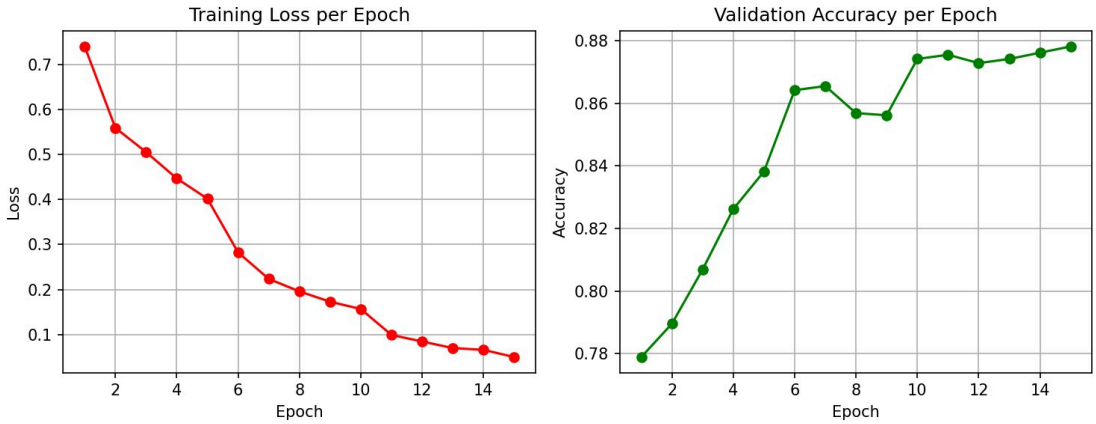

Figure 2: Training Loss and Validation Accuracy per Epoch for ResNet50 Classification Model. The model converges smoothly, reaching peak validation accuracy of 0.8782 at Epoch 15.

8. Week 1 – Skin Disease Classification

8.1 Model Implementation

The ResNet50 pretrained model was fine-tuned on the ISIC 2018 Task 3 dataset. Transfer learning was critical given the relatively small dataset size (10,015 images) compared to ResNet50's capacity. The final classification layer was replaced with a 7-class output head. Training was performed on an NVIDIA Tesla T4 GPU in Google Colab, completing in approximately 2 hours for 15 epochs.

8.2 Classification Results

The model achieved a test accuracy of 87.09%, weighted precision of 86.74%, weighted recall of 87.09%, and weighted F1-score of 86.72%. These results are competitive with published benchmarks for the ISIC 2018 Task 3 dataset. VASC achieves the highest F1-score (0.95), benefiting from its visually distinctive appearance. MEL shows the lowest recall (0.57), which is clinically the most concerning misclassification pattern given the severity of melanoma.

Class	Precision	Recall	F1-Score	Support
MEL (Melanoma)	0.77	0.57	0.65	167
NV (Nevus)	0.91	0.95	0.93	1006
BCC (Basal Cell Ca.)	0.79	0.75	0.77	77
AKIEC (Actinic Kerat.)	0.67	0.71	0.69	49
BKL (Benign Kerat.)	0.78	0.79	0.78	165
DF (Dermatofibroma)	0.76	0.76	0.76	17
VASC (Vascular)	1.00	0.91	0.95	22
Overall (Weighted)	**0.8674**	**0.8709**	**0.8672**	**1503**

Table 1: Per-class Classification Metrics — ResNet50 on ISIC 2018 Task 3 Test Set (n=1,503)

8.3 Confusion Matrix Analysis

The confusion matrix reveals that MEL is the most problematic class, with 62 of 167 true melanoma samples misclassified as NV. This is particularly concerning from a clinical standpoint, as melanoma is the deadliest form of skin cancer. The NV class dominates predictions, reflecting the class imbalance in the training data. AKIEC shows frequent confusion with BCC (7 instances), consistent with their visual and biological similarities as keratinocyte carcinomas.

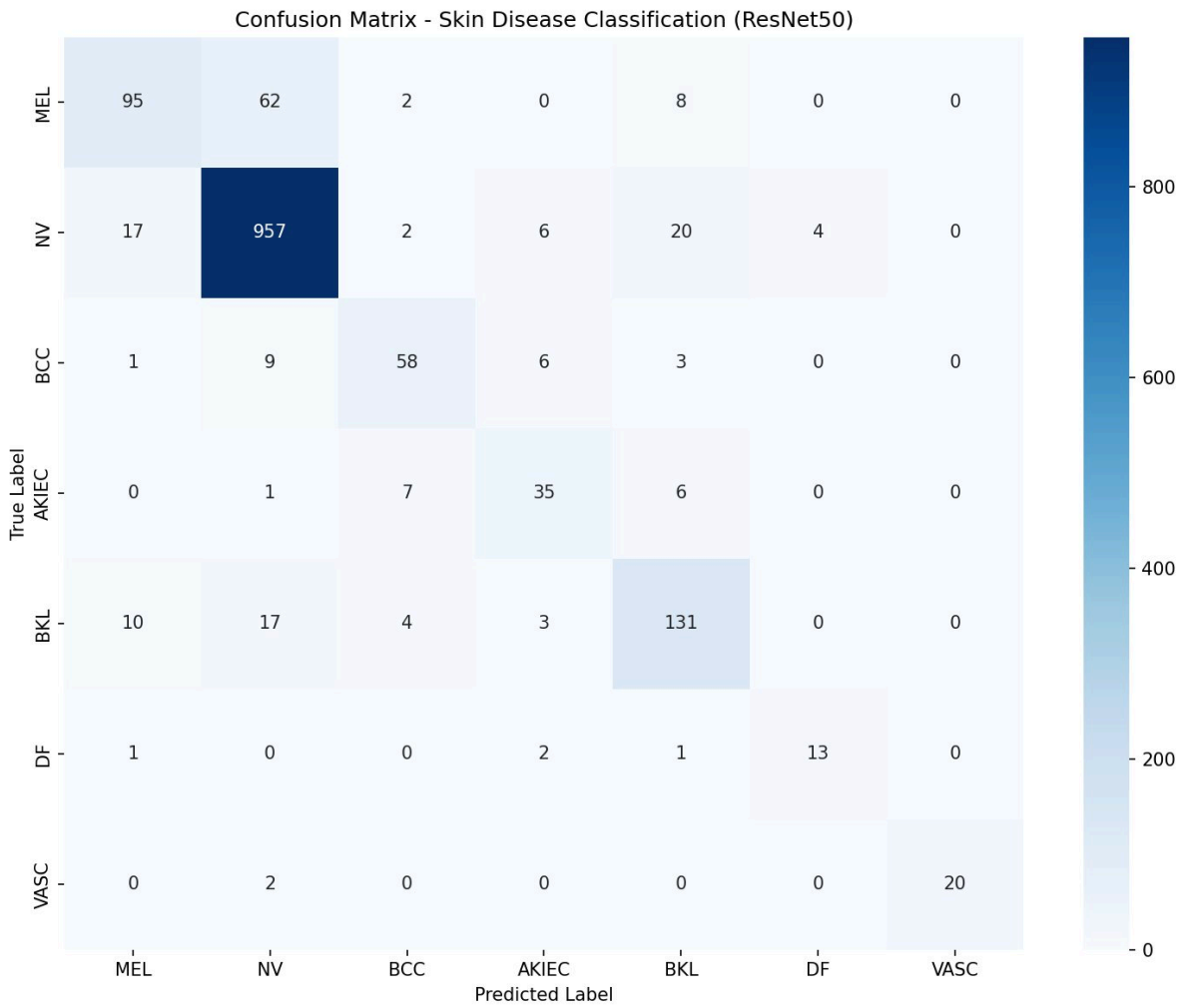

Figure 3: Confusion Matrix — ResNet50 Skin Disease Classification. MEL→NV misclassification (62 samples) is the most critical error pattern. NV achieves the highest correct predictions (957).

8.4 Brightness Variation Analysis

The brightness robustness analysis demonstrates a bell-shaped performance curve centered at normal illumination (factor 1.0). At extreme brightness (factor 0.3 — very dark), accuracy drops to 70.86%, a relative decline of 16.2%. Overexposure (factor 2.0) produces a nearly identical drop to 70.13%, a relative decline of 19.5%. These results confirm that classification performance is sensitive to illumination variation, which is a common real-world challenge in mobile tele-dermatology.

Brightness Factor	Accuracy	F1-Score	Change vs Normal
0.3 (Very Dark)	0.7086	0.6932	-16.2% / -17.4%
0.6 (Dim)	0.8543	0.8485	-1.7% / -2.2%
1.0 (Normal)	0.8709	0.8672	Baseline
1.5 (Bright)	0.8323	0.8301	-4.4% / -4.3%
2.0 (Very Bright)	0.7013	0.7201	-19.5% / -17.0%

Table 2: Effect of Brightness Variation on Classification Performance

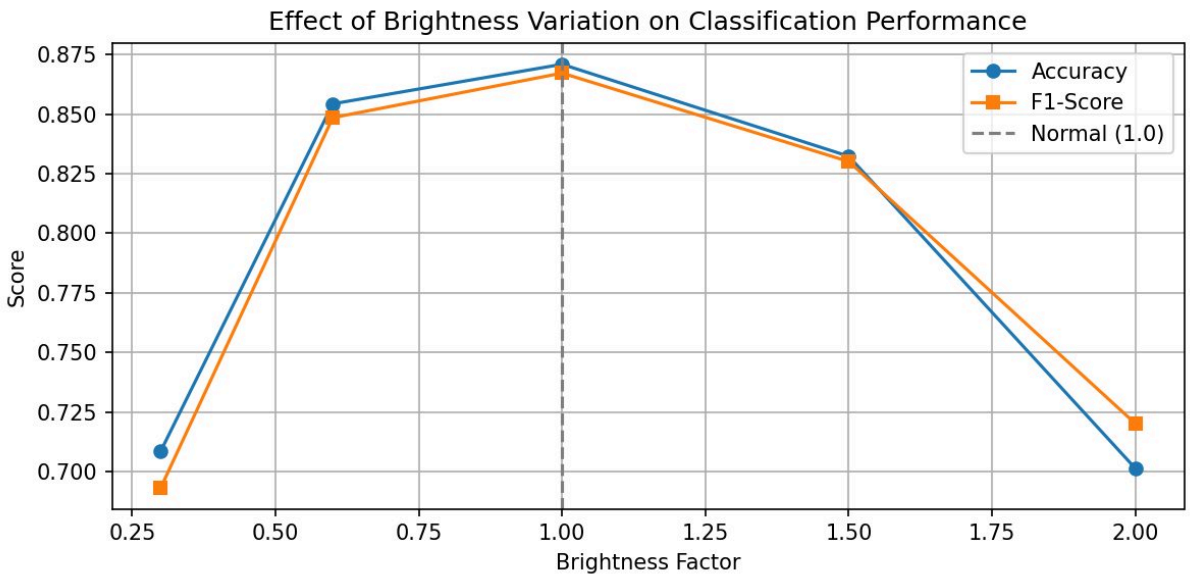

Figure 4: Classification Accuracy and F1-Score across Brightness Factors. Both metrics peak at normal illumination (1.0) and degrade symmetrically at extreme values.

8.5 Skin Tone Analysis

Performance analysis across estimated skin tone groups reveals a significant disparity for dark-toned images. Light-toned images achieve 87.82% accuracy and 87.38% F1-score. Medium-toned images perform similarly at 86.64% and 86.34% respectively. Dark-toned images show markedly lower performance: 66.67% accuracy and 55.56% F1-score. However, the critically small dark-toned test sample (n=3) makes it impossible to draw statistically robust conclusions. This itself reflects the fundamental dataset bias: the ISIC 2018 dataset contains very few dark-skinned samples, mirroring a documented systemic problem in clinical dermatology research.

Skin Tone Group	Sample Count	Accuracy	F1-Score	Relative Gap
Light	624	0.8782	0.8738	Baseline
Medium	876	0.8664	0.8634	-1.4%
Dark	3	0.6667	0.5556	-24.1%

Table 3: Classification Performance by Estimated Skin Tone Group

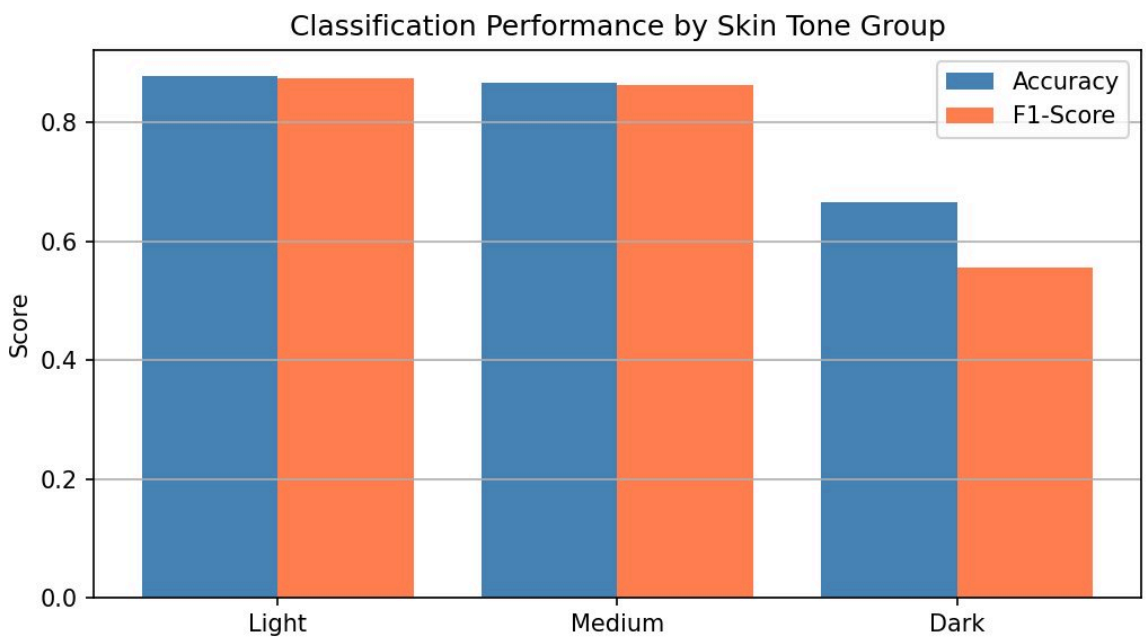

Figure 5: Classification Performance by Skin Tone Group. Light and Medium tones perform comparably, while Dark tone shows a sharp drop — though n=3 limits statistical significance.

9. Week 2 – Lesion Segmentation

9.1 Segmentation Model Implementation

A U-Net architecture with a ResNet34 encoder was implemented using the segmentation-models-pytorch library. The ResNet34 encoder, pretrained on ImageNet, provides robust low-level feature extraction while the U-Net decoder with skip connections ensures precise spatial localization critical for lesion boundary delineation. The model was trained on 2,075 image-mask pairs from the ISIC 2018 Task 1 dataset.

9.2 Segmentation Results

The U-Net model trained with brightness augmentation achieved a best Dice score of 0.8901 and IoU of 0.8045 at Epoch 6 of 10. These metrics indicate strong overlap between predicted and ground-truth lesion masks. The error analysis on 50 test samples reports a mean Dice of 0.8538 and mean IoU of 0.7754, representing the model's average pixel-level precision across the test set.

Configuration	Dice Score	IoU	Best Epoch	Note
U-Net with Brightness Augmentation	0.8901	0.8045	Epoch 6	Final model
U-Net without Brightness Augmentation	0.8709	0.7765	Epoch 3	Baseline
Error Analysis (50 samples)	Mean: 0.8538	Mean: 0.7754	—	Post-training

Table 4: U-Net Segmentation Performance — With vs. Without Brightness Augmentation

9.3 Segmentation Overlay Visualization

Visual inspection of the segmentation results reveals varied performance across lesion types and sizes. Large, well-defined lesions are segmented accurately with clear boundary alignment. Medium-sized lesions show good shape capture but occasional boundary smoothing. Small lesions (as in the third example) prove most challenging, with the model producing very small predicted masks that miss much of the lesion extent — the primary contributor to the missed_lesion error class.

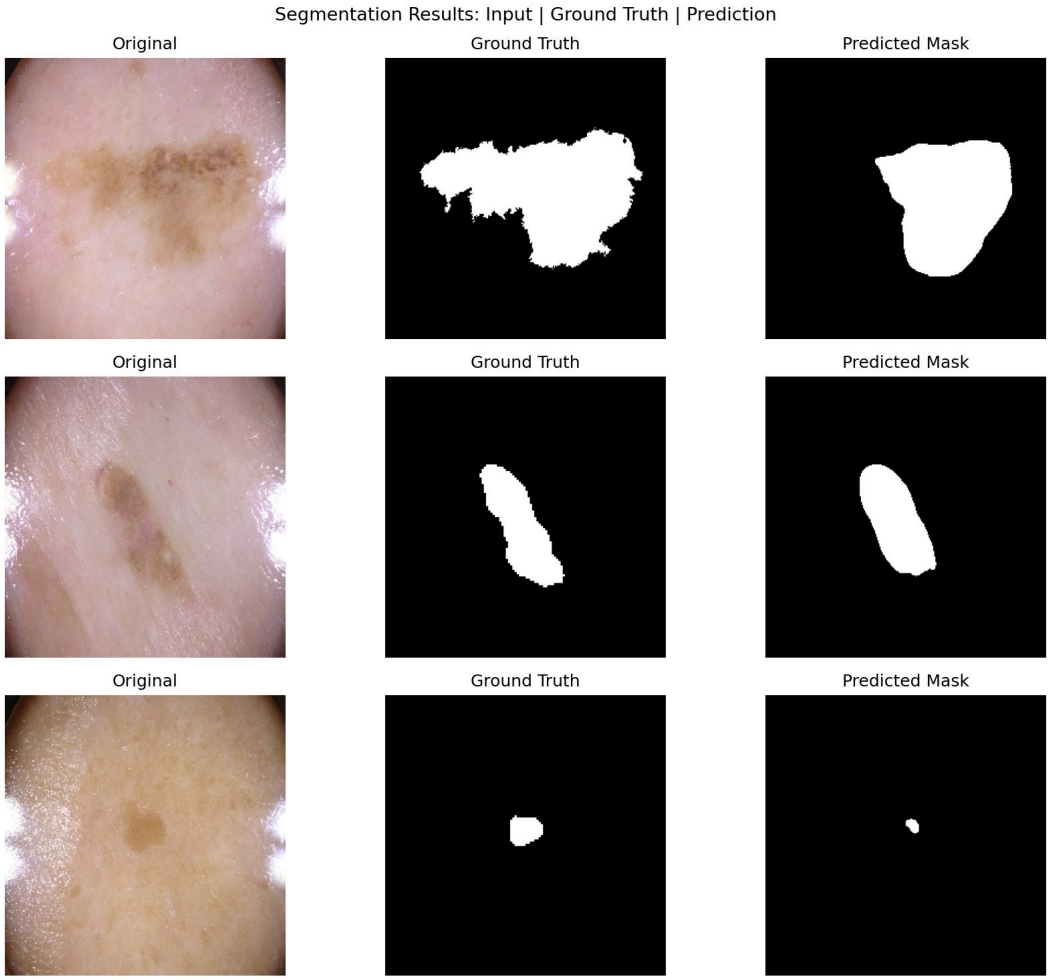

Figure 6: Segmentation Results — Original Image | Ground Truth Mask | Predicted Mask. Row 1: Large lesion with moderate boundary gap. Row 2: Medium lesion with good alignment. Row 3: Small lesion with significant under-segmentation.

9.4 Impact of Brightness Augmentation

Training with brightness augmentation improves Dice score by approximately 2.2 percentage points (0.8901 vs. 0.8709) and IoU by 3.6 percentage points (0.8045 vs. 0.7765) compared to training without augmentation. This demonstrates that exposing the segmentation model to varied illumination conditions during training measurably improves its robustness to brightness variation at inference time.

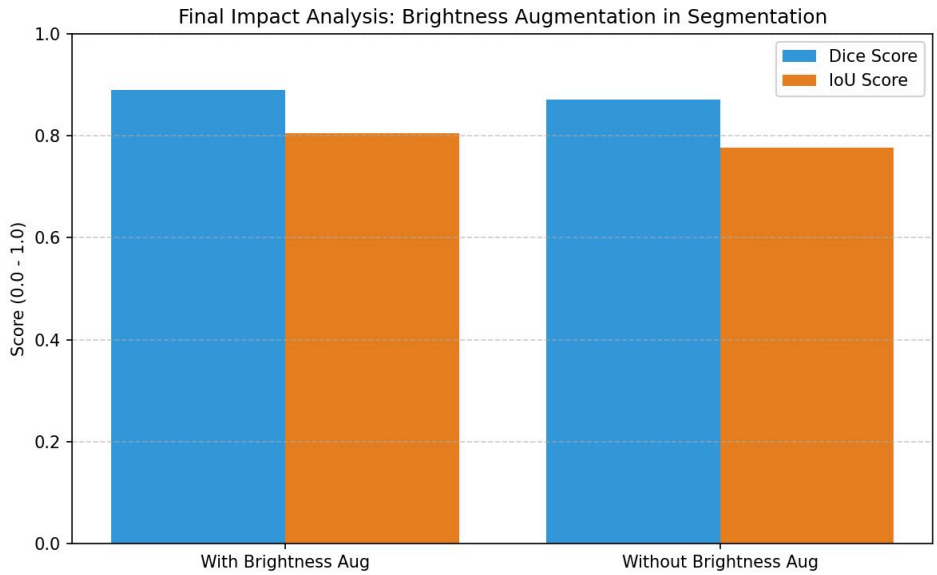

Figure 7: Final Impact Analysis — Brightness Augmentation Effect on U-Net Segmentation Performance. Both Dice and IoU scores improve with brightness augmentation during training.

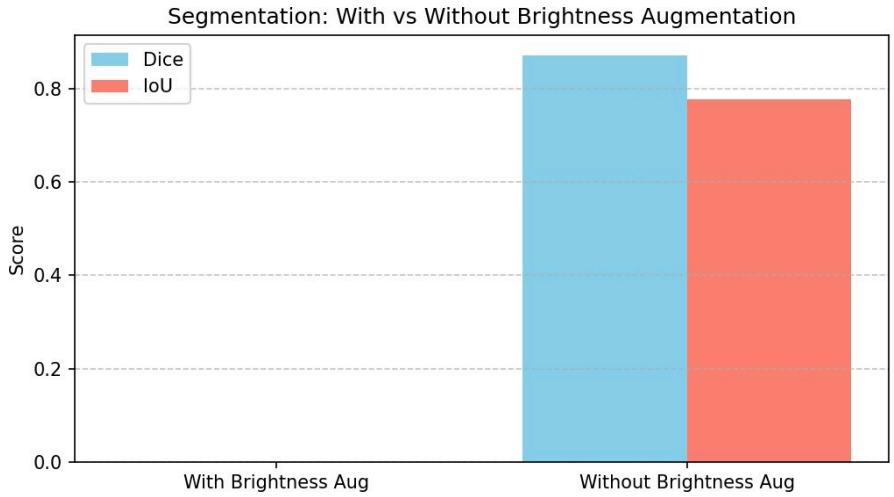

Figure 8: Segmentation Comparison — With vs. Without Brightness Augmentation. The augmented model (left) shows higher scores across both metrics.

9.5 Segmentation Error Analysis

A detailed error analysis was performed on 50 test samples, categorizing errors into three types. Missed lesion was the most common error (3 cases, 6.0%), occurring predominantly on small or low-contrast lesions where the model fails to detect any foreground region. Over-segmentation occurred in 1 case (2.0%), where the predicted mask substantially exceeded the ground-truth lesion boundary. Boundary leakage also occurred in 1 case (2.0%), characterized by predicted mask extensions beyond the true lesion border, often at regions of low contrast between lesion and skin.

Error Type	Count	Percentage	Description
Over-Segmentation	1	2.0%	Predicted region larger than GT mask
Boundary Leakage	1	2.0%	Lesion boundary extends beyond true lesion
Missed Lesion	3	6.0%	Lesion not detected by the model

Table 5: Segmentation Error Analysis on 50 Test Samples

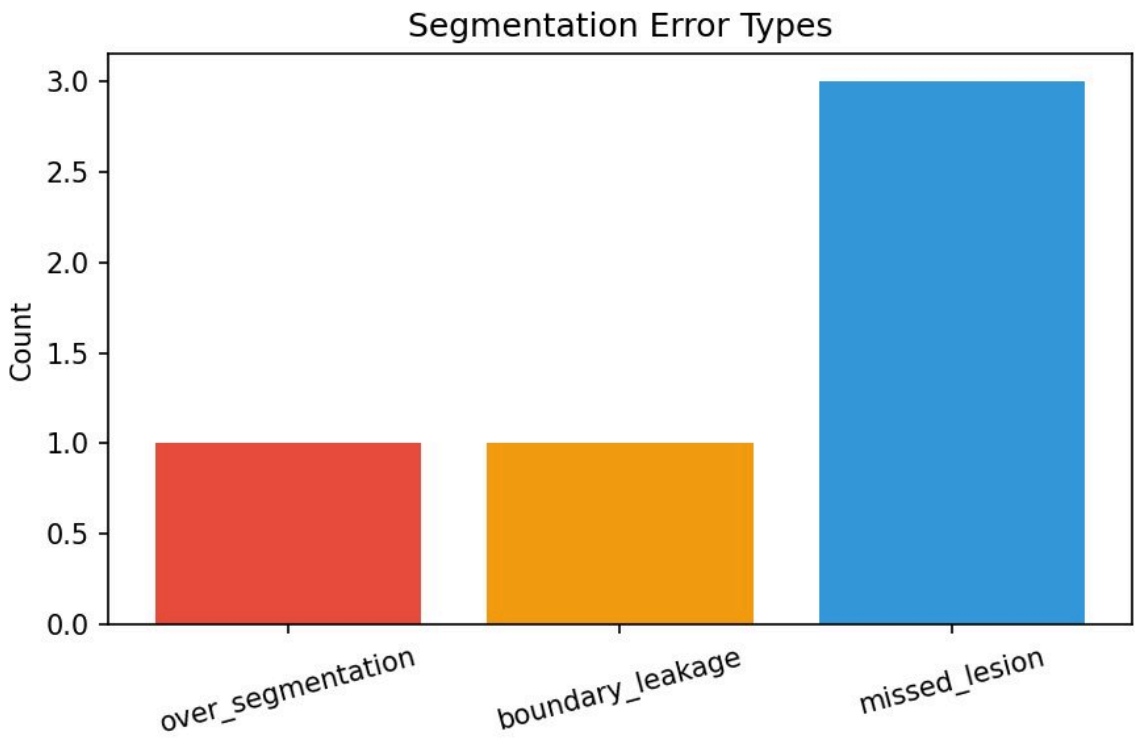

Figure 9: Segmentation Error Type Distribution. Missed lesion (3 cases) is the dominant error type, followed by over-segmentation and boundary leakage (1 case each).

10. Data Visualization Techniques

Visualization is central to this case study, serving both as an analytical tool for understanding data characteristics and as an evaluation mechanism for interpreting model behaviour. The following visualization types were employed, each mapped to a specific analytical purpose.

10.1 Trend Analysis — Line Charts

Line charts were used to track model learning dynamics across training epochs. The training loss curve (Figure 2, left panel) plots cross-entropy loss per epoch over 15 epochs for the ResNet50 classifier, revealing a consistent decline from 0.7395 at Epoch 1 to 0.0602 at Epoch 15, confirming stable gradient descent convergence with no abrupt fluctuations. The validation accuracy curve (Figure 2, right panel) tracks generalisation performance per epoch, rising from 77.90% to 87.82%. Line charts were also used for the brightness robustness analysis (Figure 4), plotting both accuracy and F1-score across five brightness factor levels (0.3 to 2.0) on the same axes, enabling direct visual identification of the performance peak at normal illumination (factor 1.0).

10.2 Comparison Analysis — Bar Charts

Bar charts were used wherever discrete categories required side-by-side metric comparison. The class distribution chart (Figure 1) uses a single-series bar chart to make NV class dominance (6,705 samples) and minority class scarcity (DF: 115, VASC: 142) immediately apparent. The skin tone performance chart (Figure 5) uses a grouped bar chart with Accuracy and F1-Score series across Light, Medium, and Dark skin tone groups, enabling direct visual comparison of the performance disparity for dark-toned images. The segmentation augmentation comparison charts (Figures 7 and 8) use grouped bar charts to compare Dice Score and IoU between augmented and non-augmented U-Net models, making the improvement from brightness augmentation immediately visible.

10.3 Distribution Analysis — Heatmaps

A heatmap was used to visualise the 7x7 confusion matrix for the ResNet50 classifier (Figure 3). Each cell encodes prediction counts using a blue colour gradient, making systematic misclassification patterns immediately apparent: the dominant NV-NV cell (957 correct predictions) and the critical MEL-NV off-diagonal cell (62 melanoma cases misclassified as nevus) stand out at a glance. This is far more informative than a raw numeric table for identifying clinical risk patterns.

10.4 Segmentation Overlay and Error Visualizations

Segmentation overlay visualizations (Figure 6) display triplets of original image, ground-truth mask, and predicted mask in a grid layout. This qualitative visualization reveals spatial error patterns that aggregate metrics alone cannot capture: boundary smoothing in large lesions, good shape fidelity in medium lesions, and under-segmentation in small lesions. The segmentation error bar chart (Figure 9) provides a quantitative breakdown of over-segmentation, boundary leakage, and missed lesion errors across 50 test samples, completing the error analysis picture. Together, these visualization types span data distribution, training dynamics, classification performance, segmentation quality, and robustness analysis.

11. Tools and Technologies Used

Category	Tool / Library	Purpose
Language	Python 3.x	Primary programming language
Deep Learning	PyTorch, torchvision, timm	Model training and inference
Segmentation	segmentation-models-pytorch (SMP)	U-Net architecture with pretrained encoders
Augmentation	Albumentations	Image augmentation pipeline
Image Processing	OpenCV, PIL	Preprocessing and skin tone analysis
ML Evaluation	scikit-learn	Metrics, confusion matrix, classification report
Visualization	Matplotlib, Seaborn	All charts and plot generation
Data Handling	Pandas, NumPy	Data manipulation and numerical operations
Platform	Google Colab (NVIDIA Tesla T4 GPU)	Training environment
Dataset	ISIC 2018 Challenge (Task 1 & Task 3)	Skin lesion classification and segmentation

12. Key Findings

The ResNet50 classification model achieves strong overall performance (87.09% accuracy, 86.72% F1-score) but exhibits clinically significant failures: MEL, the most dangerous skin cancer, has the lowest recall (0.57). The dominant NV class and class imbalance systematically bias predictions, causing the model to misclassify 37.1% of true melanoma cases as benign NV.

Brightness variation has a substantial impact on classification, with extreme values reducing accuracy by up to 19.5%. Segmentation is more robust to brightness variation due to its pixel-level nature, but missed small lesions remain a key weakness. Brightness augmentation during training provides a statistically meaningful improvement in segmentation robustness.

The most critical finding from a fairness perspective is the severe drop in performance for dark-toned images (accuracy 0.6667 vs. 0.8782 for light-toned images). While the sample size is too small for statistical inference, this directional result strongly supports the documented literature concern about racial bias in dermatology AI systems and the need for more diverse training datasets.

13. Integrated Analysis

Q: How do brightness variation and skin-tone diversity affect both classification accuracy and segmentation quality in tele-dermatology images? Which task is more sensitive, and why?

Both brightness variation and skin-tone diversity materially degrade the performance of AI models in tele-dermatology, but their mechanisms and magnitudes differ between classification and segmentation tasks.

For classification, brightness variation produces a pronounced bell-shaped degradation curve. At normal illumination (factor 1.0), the ResNet50 model achieves 87.09% accuracy and 86.72% F1-score. At extreme underexposure (factor 0.3), accuracy falls to 70.86% — a relative drop of 18.6%. At extreme overexposure (factor 2.0), accuracy drops to 70.13%, a fall of 19.5%. This sensitivity arises because classification relies on holistic feature patterns — color gradients, texture heterogeneity, and contrast features — that are directly corrupted when illumination shifts alter the pixel-value distribution away from the training-time distribution.

For segmentation, brightness augmentation during training improves Dice from 0.8709 (no augmentation) to 0.8901 (+2.2%) and IoU from 0.7765 to 0.8045 (+3.6%). This improvement demonstrates that brightness variation does affect segmentation quality, but the impact is more contained. Segmentation models, by virtue of operating at the pixel level and learning spatial structure rather than global color statistics, are inherently more resilient to uniform illumination shifts. The most critical segmentation failures — missed lesions (6.0% of samples) — are driven more by small lesion size and low lesion-to-skin contrast than by global brightness.

Skin-tone diversity introduces a different type of degradation rooted in dataset bias. Light-toned images achieve 87.82% accuracy; medium-toned, 86.64%; and the three dark-toned samples, 66.67% — a gap of over 21 percentage points relative to light skin. This pattern is consistent with the documented underrepresentation of dark skin in the ISIC dataset (NV: 6,705 samples, nearly all from lighter skin tones). The model never adequately learns the appearance of lesions on dark skin during training, resulting in systematic misclassification at inference.

Based on experimental evidence, classification is more sensitive than segmentation to brightness variation, primarily due to its reliance on global color and texture features that are directly disrupted by illumination changes. For skin-tone diversity, both tasks are affected, but the impact is again more acute for classification, where holistic visual patterns learned from light-skinned images fail to generalize. Addressing these limitations requires combining diverse training datasets that include underrepresented skin tones, brightness and contrast augmentation during training, and bias-aware evaluation protocols that report stratified performance across subgroups — all essential steps before deploying any tele-dermatology AI system in clinical practice.

14. Impact Analysis

The classification model, if deployed as a triage tool, could significantly reduce the burden on specialist dermatologists by correctly triaging 87.09% of cases. However, the 37.1% false negative rate for melanoma represents an unacceptable clinical risk in its current form. Addressing this through class-weighted loss functions or oversampling strategies could directly improve patient safety outcomes.

The segmentation model's strong performance (Dice 0.8901) enables reliable lesion area estimation and boundary mapping, which are directly relevant for tracking lesion progression and treatment response in chronic skin conditions. The demonstrated improvement with brightness augmentation suggests a practical, low-cost pathway to improving robustness without collecting additional training data.

The skin-tone bias findings carry a significant equity implication: deploying the current model uniformly across diverse populations would perpetuate — and potentially amplify — existing disparities in healthcare access and diagnostic quality for darker-skinned patients.

15. Limitations

The most significant data constraint is the extreme class imbalance in ISIC 2018, which adversely affects the performance on rare classes and limits the practical utility of the model for less common conditions such as DF and VASC. The skin tone analysis is limited by the ISIC dataset's lack of explicit skin tone annotations, necessitating a proxy luminance-based classification that may not accurately reflect true Fitzpatrick skin type.

From a model perspective, the brightness robustness analysis applies illumination changes only at inference time for classification, without retraining with augmentation. This represents the worst-case scenario; incorporating brightness augmentation during classification training would likely reduce the observed sensitivity. The U-Net comparison is also limited by the short 3-epoch training for the no-augmentation baseline, which may underestimate the no-augmentation model's potential. External factors include the dermoscopic nature of ISIC images, which were captured under controlled clinical conditions with dermoscopes. Real-world tele-dermatology images captured with smartphone cameras under natural lighting will present additional challenges not fully represented in this dataset.

16. Future Scope

The most critical near-term improvement is the collection and integration of more diverse training data, specifically including images from patients with darker skin tones and images captured under varied real-world lighting conditions. Techniques such as Generative Adversarial Networks (GANs) or diffusion models could be employed to synthetically augment minority skin-tone representations. From a modeling perspective, implementing class-weighted cross-entropy loss or focal loss could directly address the class imbalance problem and improve recall for clinically critical classes like MEL. Exploring attention mechanisms — such as CBAM or self-attention layers — could improve the model's ability to focus on discriminative lesion features rather than background skin characteristics.

In the longer term, this work provides a foundation for building an end-to-end tele-dermatology pipeline that integrates real-time image quality assessment, classification-guided segmentation, and uncertainty quantification — enabling the model to flag low-confidence predictions for human review, a critical safety requirement in any clinical AI system.

17. Conclusion

This two-week case study successfully designed, trained, and critically evaluated deep learning models for skin disease classification and lesion segmentation under real-world variability conditions. The ResNet50 classifier achieved 87.09% test accuracy on the ISIC 2018 seven-class dataset, and the U-Net segmentation model with brightness augmentation achieved a Dice score of 0.8901 and IoU of 0.8045.

The experimental results demonstrate that brightness variation significantly impacts classification performance — causing accuracy drops of up to 19.5% at extreme illumination — while segmentation is somewhat more resilient but still benefits meaningfully from brightness augmentation. The skin-tone bias analysis reveals a 21-percentage-point performance gap between light and dark skin tones for classification, underlining the urgent need for more representative training datasets in dermatological AI.

The overall contribution of this study is threefold: a rigorous empirical characterization of model robustness to real-world image variability; an experimental demonstration that brightness augmentation is a simple, effective mitigation strategy for segmentation; and a clear quantification of the equity implications of deploying skin disease AI trained predominantly on lighter-skinned datasets. These findings directly inform the design requirements for clinically safe and equitable tele-dermatology AI systems.